

Web Search: Overview & Challenges

Information Retrieval

Sean MacAvaney

2026

Lecture Outline

- **Queries on the Web**
- Web Dynamics
- Ranking in Web search
- Link Analysis
 - PageRank
 - HITS

Taxonomy of Web Search [Broder 2002]

There are **three main classes** of queries:

- **Navigational queries:** to reach a particular site that the user has in mind (aka known-item search)
 - Reach a particular webpage/URL
- **Informational queries:** to acquire some information assumed to be present on one or more webpages.
 - Reading/bookmarking/printing are the only further user's actions
- **Transactional queries:** to reach a site where further interaction will happen (e.g. shopping, downloading, gaming)
 - User further engages/interacts with websites in the results list

It is often hard to infer intent from a query

You could imagine all three intents for this query:



The image is a screenshot of a Google search interface. At the top, the Google logo is on the left, and navigation links for Web, Images, Video, News, Maps, Desktop, and more » are on the right. Below the logo is a search bar containing the text 'doubleclick'. To the right of the search bar is a 'Search' button and links for 'Advanced Search' and 'Preferences'. Below the search bar, the 'Web' tab is selected. A blue speech bubble with the text 'One-box results' points to the search results. The results section is titled 'News results for doubleclick - View today's top stories'. It features a small icon of a newspaper and three news snippets: 'Schmidt Scoffs at Antitrust Concerns About DoubleClick Deal - PC Magazine - 10 hours ago', 'Google DoubleClick: The REAL big time payoff - ZDNet - 2 hours ago', and 'Google's Schmidt defends DoubleClick ads deal - Hollywood Reporter - Apr 17, 2007'. Below the news results is a section titled 'DoubleClick: Digital Advertising'. It contains a paragraph: 'DoubleClick enables agencies, marketers and publishers to work together successfully and profit from their digital marketing investments.' followed by a link to 'www.doubleclick.com/' with additional information: '- 22k - Apr 17, 2007 - Cached - Similar pages'. Below this are several links: 'Contact Us - www.doubleclick.com/us/contact_us/', 'Products - www.doubleclick.com/us/products/', 'DART for Publishers - www.doubleclick.com/us/products/dfp/', 'About DoubleClick - www.doubleclick.com/us/about_doubleclick/', and 'More results from www.doubleclick.com »'.

Google

Web Images Video News Maps Desktop more »

doubleclick Search Advanced Search Preferences

Web

One-box results

News results for **doubleclick** - View today's top stories

 Schmidt Scoffs at Antitrust Concerns About **DoubleClick** Deal - PC Magazine - 10 hours ago
Google **DoubleClick**: The REAL big time payoff - ZDNet - 2 hours ago
Google's Schmidt defends **DoubleClick** ads deal - Hollywood Reporter - Apr 17, 2007

DoubleClick: Digital Advertising

DoubleClick enables agencies, marketers and publishers to work together successfully and profit from their digital marketing investments.

www.doubleclick.com/ - 22k - Apr 17, 2007 - [Cached](#) - [Similar pages](#)

[Contact Us](#) - www.doubleclick.com/us/contact_us/
[Products](#) - www.doubleclick.com/us/products/
[DART for Publishers](#) - www.doubleclick.com/us/products/dfp/
[About DoubleClick](#) - www.doubleclick.com/us/about_doubleclick/
[More results from www.doubleclick.com »](#)

Queries on the Web

- Commercial web search engines do not disclose their search logs
- However, there have been a number of research studies/reports showing that:
 - Queries: 20% (navigational), 48% (informational), 30% (transactional) [Broder 2002]
 - Bing also reported in 2011 that ~30% queries are navigational
 - 33% of a user's queries are repeat queries; 87% of the time the user would click on the same result [Teevan et al., 2005]
 - Up to 40% of queries are re-finding [Teevan et al., 2007]
 - The length of queries has increased from 2.4 (2001) to 3.08 (2011) [Taghavi et al., 2011]
 - 16% of queries are ambiguous [Song et al., 2009]
 - 12%-16% of queries have local intent [Gravano et al., 2003]

In 2013, Google responded to 2 trillion+ queries

What Makes Web Search Difficult?

Size



Diversity



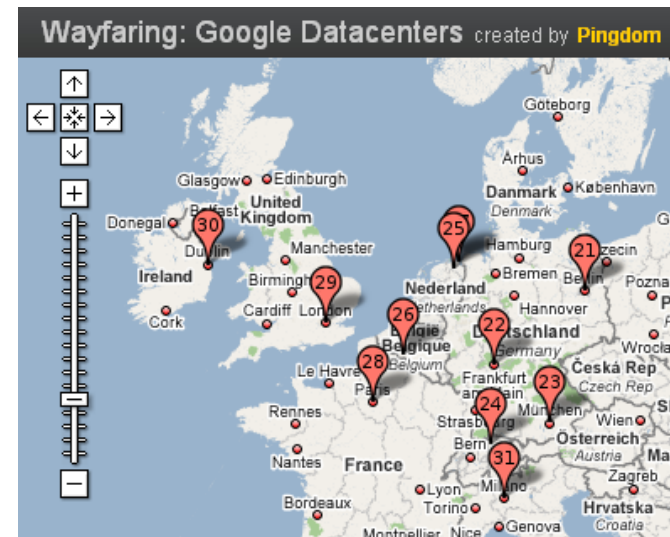
Dynamicity



- All of these three characteristics can be observed in:
 - The Web (in 2013, Google reported it has indexed 30 trillion pages)
 - Web users (estimated at ~2 billion at the end of 2012; ~4.3 billion users in early 2019)

Web Search Engine Costs

- Quality and performance requirements imply large amounts of compute resources, i.e., very large data centers
- Hundreds of thousands of computers; Heavy consumption of energy:
 - Between 1.1% and 1.5% of the world's total energy use in 2010
 - Still ~1% in 2018 (following efforts in energy consumption optimization)
 - One Google search ≈ 1 KJ $\approx$ turning on a 60W light bulb for ~ 17 secs



Actors in Web Search

- **User's perspective:** accessing information
 - Relevance
 - Speed
- **Search engine's perspective:** monetisation
 - Attract more users
 - Increase the ad revenue
 - Reduce the operational costs
- **Advertiser's perspective:** publicity
 - Attract more users, c.f. Search Engine Optimisation
 - Pay little



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- **Web Dynamics**
- Ranking in Web search
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Web Dynamics

- Web is *changing* over **time** in many aspects, e.g., size, content, structure and how it is accessed by user interactions or queries
 - **Size**: web pages are added/deleted at all time
 - **Content**: web pages are edited/modified
 - **Query**: users' information needs change
 - **Usage**: users' behaviour change over time

Implications: Crawling, Indexing, Ranking

Temporal Web Dynamics

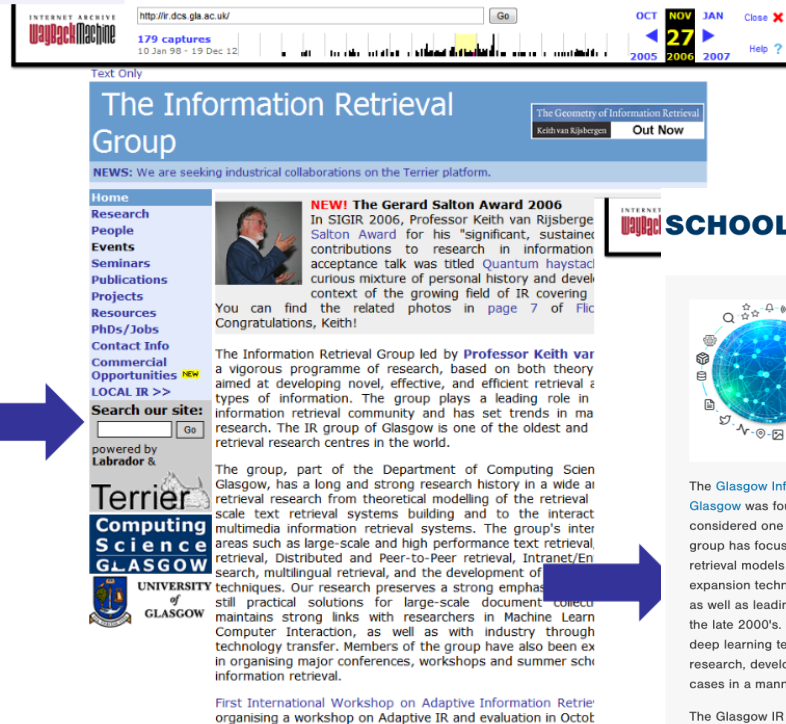
- **Time** is pervasive in information systems
 - New documents appear all the time
 - Document content changes over time
 - Query volume changes over time (e.g. lower on week-ends)
- **What's relevant** to a query changes over time
 - E.g., U.S. Open 2019 (in June->Golf vs. Sept->Tennis)
 - E.g., U.S. Open 2019 (before, during, after event)
- **User interaction** changes over time
 - E.g., anchor text, “likes”, query-click streams, social networks
- **Relations** between entities change over time
 - E.g., President of the U.S. is $\triangleleft$ [in 2020 vs. 2012]

**Temporal aspects are of considerable importance
in Web Search**

Content Dynamics



1998



2006

34% pages do not change [Adar et al., 2009]

- avg change among rest: every 123 hours

- popular pages change more often

Previous degree of content change a good predictor of future change [Fetterly et al. 2003]

SCHOOL OF COMPUTING SCIENCE



The Glasgow Information Retrieval Group within the School of Computing Science at the University of Glasgow was founded 32 years ago in 1986 by Professor C. J. 'Keith' van Rijsbergen, often considered one of the founders of modern Information Retrieval (IR). From its outset, the Glasgow IR group has focused on improving the effectiveness of IR systems, inventing new logic & probabilistic retrieval models in the 90's and early 2000's, followed by the development of adaptive query expansion techniques, interactive multimedia models, the Divergence From Randomness framework, as well as leading research into quantum, expertise search and search result diversification models in the late 2000's. Since then, the Information Retrieval group embraced emerging machine learning and deep learning technologies for very large corpora and data streams, and have been at the forefront of research, development and application of those technologies for search and recommendation use-cases in a manner that ensures both effectiveness and efficiency.

The Glasgow IR Group has a strong research track record. Indeed, the ACM Digital Library shows that the group is **ranked first** by number of papers (429) at the SIGIR conference (the top CORE A* conference in the IR field). Meanwhile, a recent **study** by Microsoft Research of the 40 years of SIGIR showed the University of Glasgow as the 5th most cited university at the conference and the 1st in Europe. The group is also renowned for developing the popular open source IR platform, **Terrier.org**, which has been downloaded over 60,000 since its first release in 2004 and is cited by over **3500 research papers**. Furthermore, the group has a long history of engagement with the public and industry sectors from small SMEs to multinational corporations.

The Informer magazine of BCS's Information Retrieval Specialist Group carried a **recent profile** on the Glasgow Information Retrieval Group.

+ Topics

+ Projects

2023

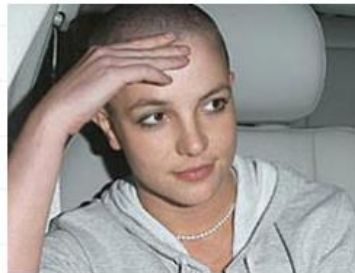
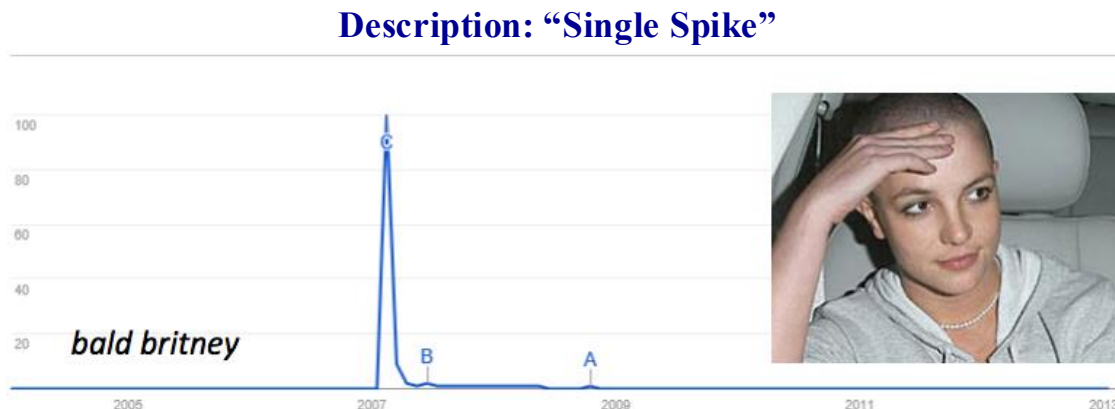
14

Query Dynamics

- Search queries exhibit temporal patterns – Spikes or seasonality



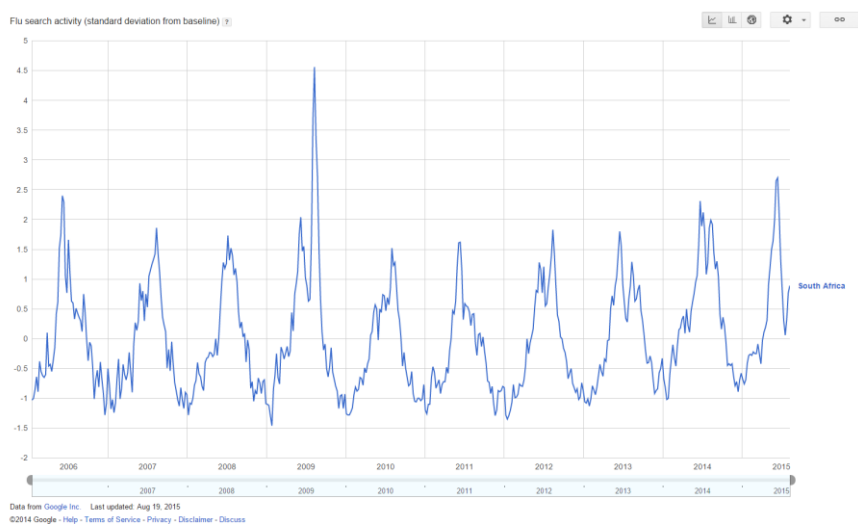
Challenges: resolving ambiguities (e.g. us open) and demoting older pages



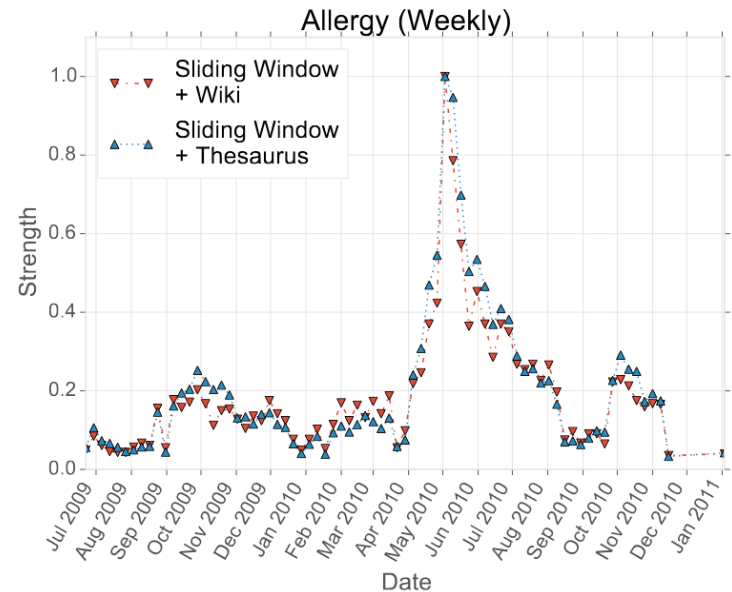
Challenges: Detecting spikes, crawling and ranking fresh documents (often news)

Query Dynamics

Spikes in mentions can be used to infer things about the real world
e.g., disease outbreaks



https://commons.wikimedia.org/wiki/File:Google_Flu_Trends_Data,_South_Africa.png



[Yates et al 2014]

Temporal Reformulations

Users' information needs change over time

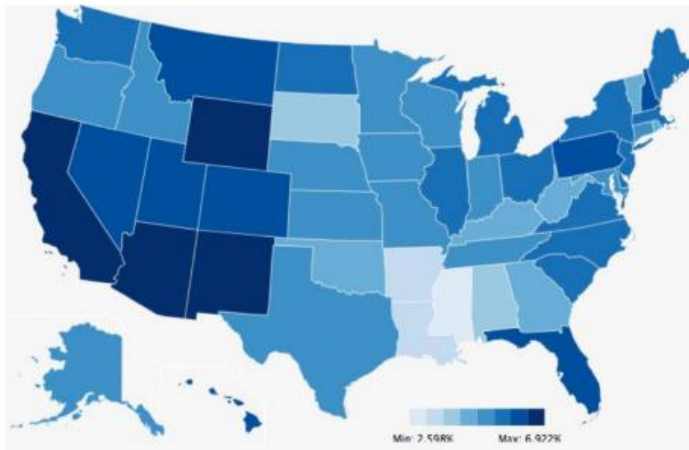
Jan07	Feb07	Mar07	Apr07	May07	Jun07
pumpkin patch	pumpkin seeds	--	pumpkin seeds	growing pumpkins	growing pumpkins
--	pumpkin pictures	--	pumpkin patch	pumpkin faces	pumpkin
--	--	--	--	pumpkin carving	--



Jul07	Aug07	Sep07	Oct07	Nov07	Dec07
growing pumpkins	growing pumpkins	pumpkin pictures	pumpkin pictures	pumpkin pictures	growing pumpkins
--	pumpkin pictures	pumpkin carving patterns	pumpkin carving patterns	pumpkin recipes	--
--	pumpkin decorating	pumpkin decorating	pumpkin decorating	growing pumpkins	--

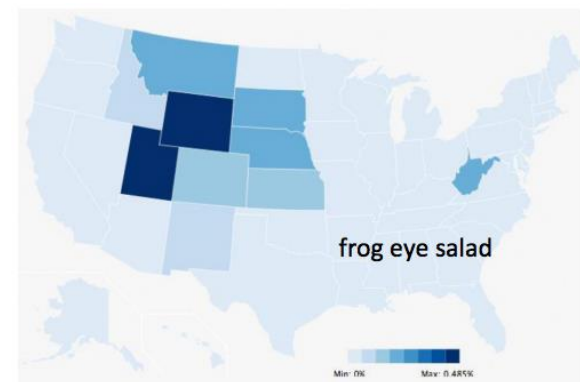
Location in Web Search

- Query volume varies across different **locations**
- Query popularity varies differently in different **locations**
- The meaning of a query could be different in different **locations**



turkey

Frequency distribution of different queries during thanksgiving



frog eye salad



millionaire pie

Location Ambiguity



Figure 7.11: Map of which location you are most likely to mean when referring to “Cambridge” in English dependent on your current location: Cambridge, UK (red), Cambridge, Massachusetts (green), Cambridge, New Zealand (blue)

Source: Overell 2009

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Ranking in Web Search

Given a **corpus** and a **query**, **rank** relevant webpages (documents) before non-relevant webpages

- **corpus**: a subset of indexed webpages
- **query**: short keyword query and any other **user/context data**
- **rank**: document scoring function

Goal is to estimate: $p(r \mid d, x)$

r relevance; d document; x **context** (e.g. query, user, location)

Ranking in Web Search

$$p(r \mid d, x) \propto s(f_i, M)$$

- f_i : set of ranking features for document i
- M : model parameters
- s : document scoring function (typically a learning to rank method)

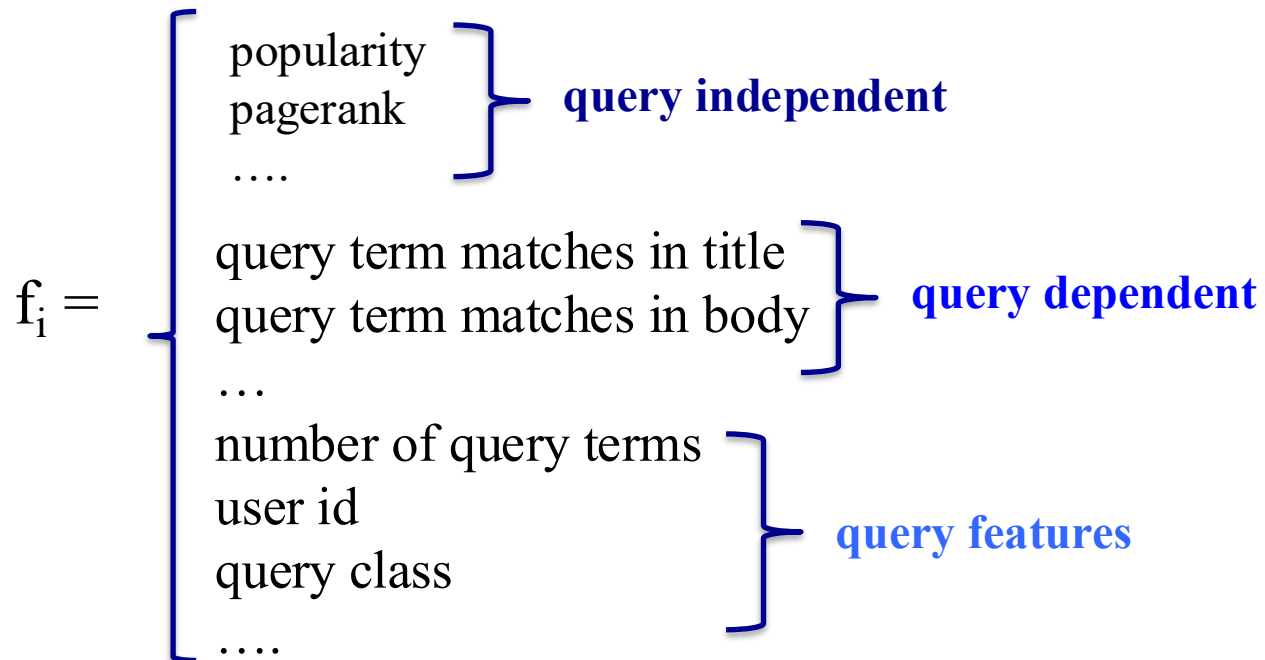
$$f_i = \left\{ \begin{array}{l} \text{query term matches in title} \\ \text{query term matches in body} \\ \dots \\ \text{document length} \\ \text{popularity} \end{array} \right.$$

Ranking in Web Search

- The recent history of web search has focused on:
 - Developing better ranking **features** (f_i)
 - Improving the optimisation of model **parameters** (M)
 - Exploring alternative **learning to rank** methods (s)

End-to-end deep learning architectures (e.g. dense retrieval approaches) are also increasingly emerging

Recall: Ranking Features



Learning to rank models determine the importance and combination of features (See Lecture 8)

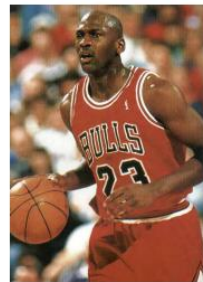
Ranking Features

- Important to select features likely to be **correlated** with relevance
 - Term matching scores (e.g. BM25, language model, PL2)
 - Field matching scores (e.g. PL2F)
- Web environment invites **unique** features:
 - **Link structure**
 - **User signals**
 - **Dynamics**

User Signals - Personalization

- Relevance of a document depends on the user's **context** (e.g. time, location, demographics)
- Users have **different interests** (e.g. sports) which are reflected in their **short** and **long-term** search history
- Queries could be **ambiguous** for the search engine; **personalisation** signals help to resolve that

Which Jordan?



User Signals - Personalization

The image shows a Google search interface with the query 'trec'. The top search bar has a yellow highlight on the 'Sign in' button. Below the search bar, the 'Web' tab is selected. The search results show 'About 4,010,000 results (0.18 seconds)'. The first result is 'TREC - Home Page' from 'www.trec.state.tx.us/'. Below this, there are links for 'Contract Forms', 'Complaints', 'Online Services', and 'Licensee and Registrant Data'. A yellow highlight is on the 'Online Services' link. Below the search results, there is a section for 'Text REtrieval Conference (TREC) Home Page' from 'trec.nist.gov/'. Below this, there is a section for 'TREC Session Track 2013 - University of Delaware' from 'ir.cis.udel.edu/sessions/'. A yellow highlight is on the 'TREC Session Track 2013 - University of Delaware' section. The bottom of the page shows a yellow highlight on the text 'You've visited this page 2 times. Last visit: 5/19/13'.

Google trec Sign in

Web Images Maps Shopping More Search tools

About 4,010,000 results (0.18 seconds)

TREC - Home Page
www.trec.state.tx.us/ ▼
Official site of the Texas Real Estate Commission, the body that governs real estate practices in the state. Includes licensing information, laws, contact ...

[Contract Forms](#)
Promulgated Contract Forms, Addenda, Amendments.

[Complaints](#)
Complaints. The procedures for filing a complaint, as well as ...

[Online Services](#)
Click the "My License Services" yellow button

[Licensee and Registrant Data](#)
Licensee and Registrant Data to Download. Looking for ...

More results from trec.state.tx.us »

Google trec ladh Share

Web Images Maps Shopping More Search tools

About 4,010,000 results (0.19 seconds)

TREC - Home Page
www.trec.state.tx.us/ ▼
Official site of the Texas Real Estate Commission, the body that governs real estate practices in the state. Includes licensing information, laws, contact ...

[Contract Forms](#)
Promulgated Contract Forms, Addenda, Amendments.

[Licensee and Registrant Data](#)
Licensee and Registrant Data to Download. Looking for ...

[Online Services Status](#)
Click the "My License Online Services" yellow button below ...

[Complaints](#)
Complaints. The procedures for filing a complaint, as well as ...

More results from trec.state.tx.us »

Text REtrieval Conference (TREC) Home Page
trec.nist.gov/ ▼
An annual information retrieval conference and competition, the purpose of which is to support and further research within the information retrieval community.

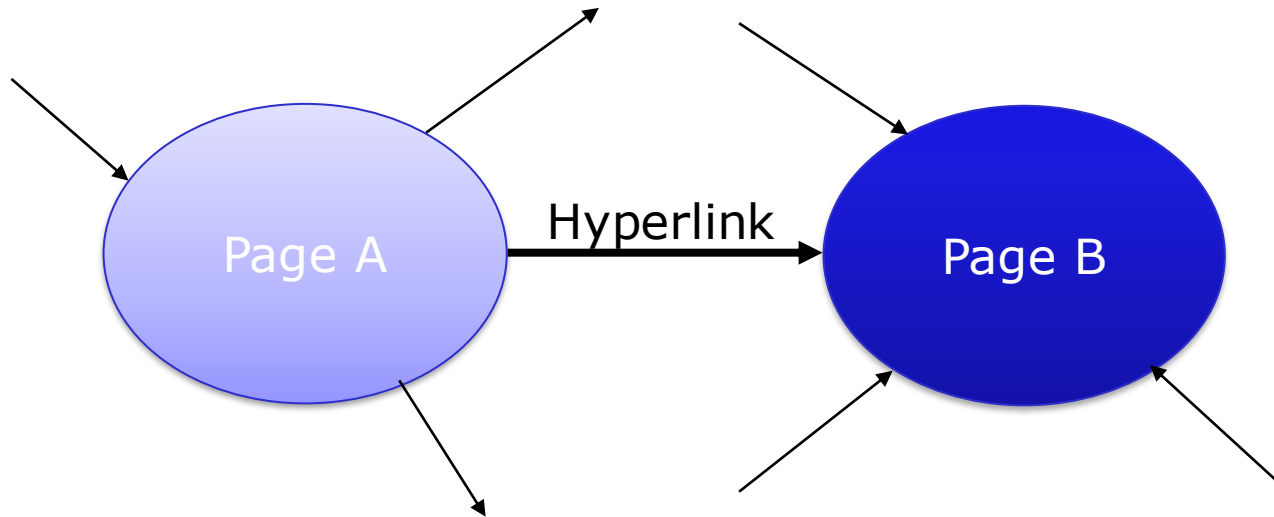
TREC Session Track 2013 - University of Delaware
ir.cis.udel.edu/sessions/ ▼
10+ items - The Session track is running for a fourth time in 2013.
Mailing list trec-sessions@udel.edu.
Subscribe at https://udel.edu/mailman/listinfo/trec-sessions.

You've visited this page 2 times. Last visit: 5/19/13

Query Independent Features

- There is a lot of **junk** on the web (e.g. spam, irrelevant forums)
- Knowing what users are **reading** is a valuable source for knowing what is not junk
- Ideally, we would be able to monitor everything the user is reading and use that information for ranking; this is achieved through toolbars, browsers, operating systems, DNS
- In 1998, no search companies had browsing data.
How did they address this lack of data?

Basic Concepts



Intuition 1: A hyperlink connotes a conferral of authority on page B, by the author of page A (quality signal) [**means of recommendation**]

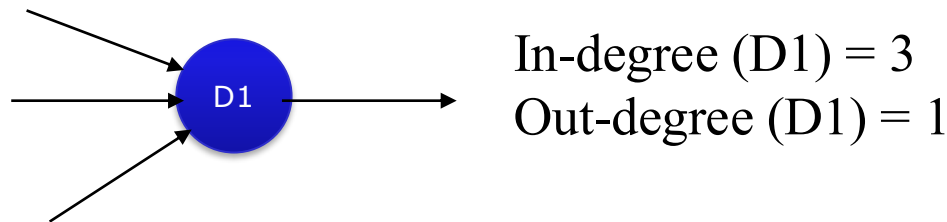
Intuition 2: The anchor of the hyperlink describes the target page (textual context) [**means of content enhancement**]

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Link Analysis

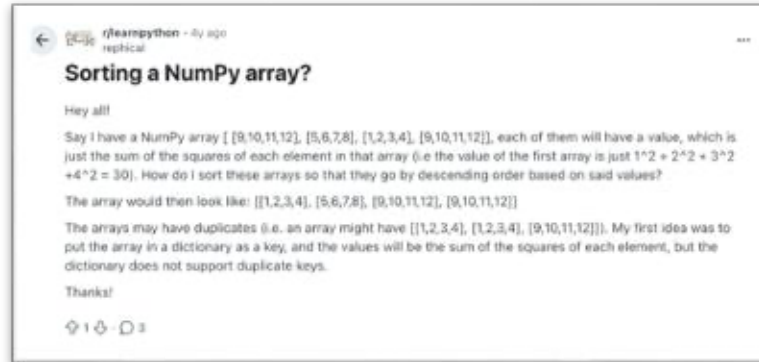
- **Link-based ranking:** Use hyperlinks to rank web documents
 - Use link counts as simple measures of popularity/authority
 - In-degree: counts the number of in-links to a given webpage



- The 2nd generation web search engines ranked webpages by combining:
 - **Content-based evidence:** e.g. scores obtained using vector space model, probabilistic model, etc.
 - **Authoritativeness:** link-based ranking

Main Idea: Wisdom of the Crowd

Only a few pages
link to this one



Worse page?

https://www.reddit.com/r/learnpython/comments/rr7s5c/sorting_a_numpy_array/

Lots of pages
link to this one

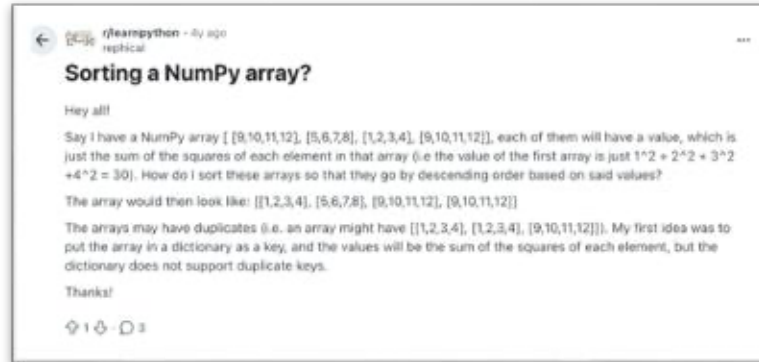


Better page?

<https://stackoverflow.com/questions/5540148/how-to-sort-an-integer-array-in-place-in-python>

Main Idea: Wisdom of the Crowd

A few **good** pages
link to this one



Better page?

https://www.reddit.com/r/learnpython/comments/rr7s5c/sorting_a_numpy_array/

spam

spam

spam

Lots of **bad** pages
link to this one



Worse page?

spam

spam

spam

<https://stackoverflow.com/questions/5540148/how-to-sort-an-integer-array-in-place-in-python>

PageRank: Random Surfer Model

[Brin and Page 1998]

- Claimed to be less susceptible to **link spam** than simpler link analysis (e.g. in-degree)
- Simulates a *very large* number of users **browsing** the *entire* web
- Let users browse randomly. This is a naïve assumption but works okay in practice
- Observe how often pages get visited
- The **authoritativeness** of a page is a function of its popularity in the simulation

PageRank (Simple Concepts)

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

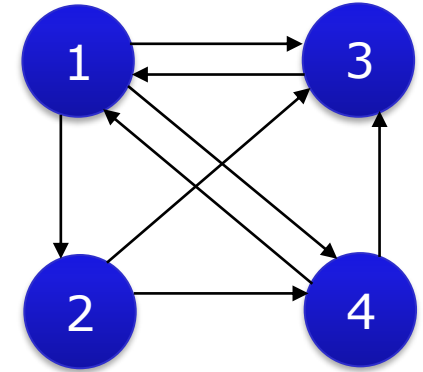
Adjacency Matrix

$$T = \begin{bmatrix} 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 1/2 & 1/2 \\ 1 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \end{bmatrix}$$

Transition Matrix

$$M = \begin{bmatrix} 0 & 0 & 1 & 1/2 \\ 1/3 & 0 & 0 & 0 \\ 1/3 & 1/2 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

$$M = [T]^T$$



$M_{ij} = 0$ if there is no link between page j and i

$M_{ij} = \frac{1}{n_j}$ otherwise, with n_j the number of outgoing links of page j

PageRank Computation

$$x_1 = \frac{x_3}{1} + \frac{x_4}{2}$$

$$x_2 = \frac{x_1}{3}$$

$$x_3 = \frac{x_1}{3} + \frac{x_2}{2} + \frac{x_4}{2}$$

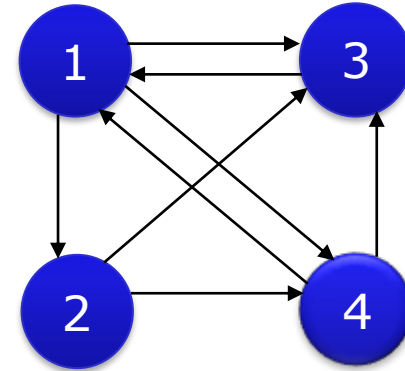
$$x_4 = \frac{x_1}{3} + \frac{x_2}{2}$$



$$\mathbf{M} \cdot \mathbf{x} = \mathbf{x}$$

where

$$\mathbf{M} = \begin{bmatrix} 0 & 0 & 1 & 1/2 \\ 1/3 & 0 & 0 & 0 \\ 1/3 & 1/2 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$



The scores can be obtained by the **power iteration** method

$\mathbf{x} = \lim_{k \rightarrow \infty} \mathbf{M}^k \mathbf{x}_0$ where $\mathbf{x}_0$ is some initial column vector with non-zero entries.

PageRank Computation

$$x_1 = \frac{x_3}{1} + \frac{x_4}{2}$$

$$x_2 = \frac{x_1}{3}$$

$$x_3 = \frac{x_1}{3} + \frac{x_2}{2} + \frac{x_4}{2}$$

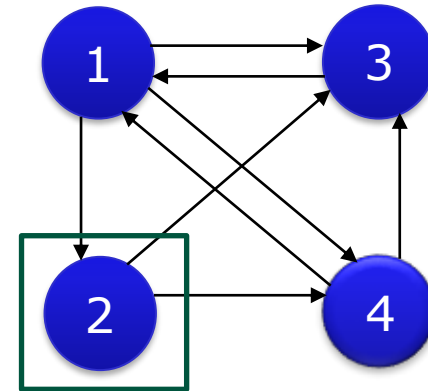
$$x_4 = \frac{x_1}{3} + \frac{x_2}{2}$$



$$\mathbf{M} \cdot \mathbf{x} = \mathbf{x}$$

where

$$\mathbf{M} = \begin{bmatrix} 0 & 0 & 1 & 1/2 \\ 1/3 & 0 & 0 & 0 \\ 1/3 & 1/2 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

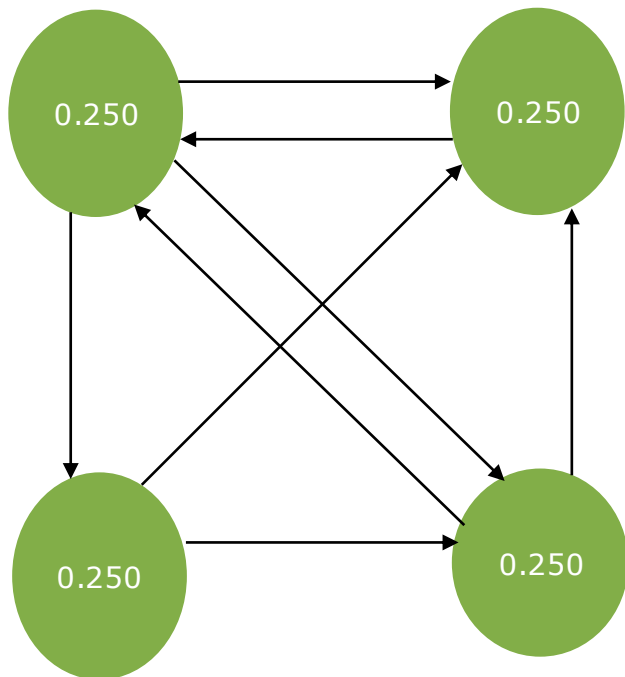


x_2 is liked to only by x_1 ,
which has a weight of $1/3$

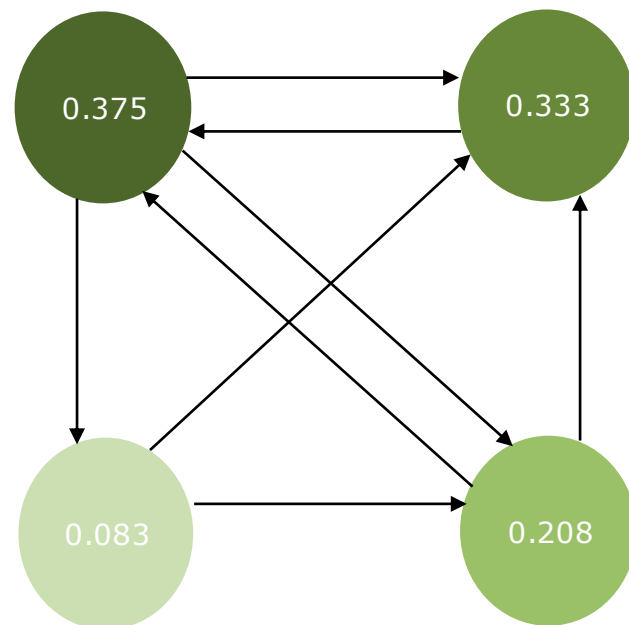
The scores can be obtained by the **power iteration** method

$\mathbf{x} = \lim_{k \rightarrow \infty} \mathbf{M}^k \mathbf{x}_0$ where $\mathbf{x}_0$ is some initial column vector with non-zero entries.

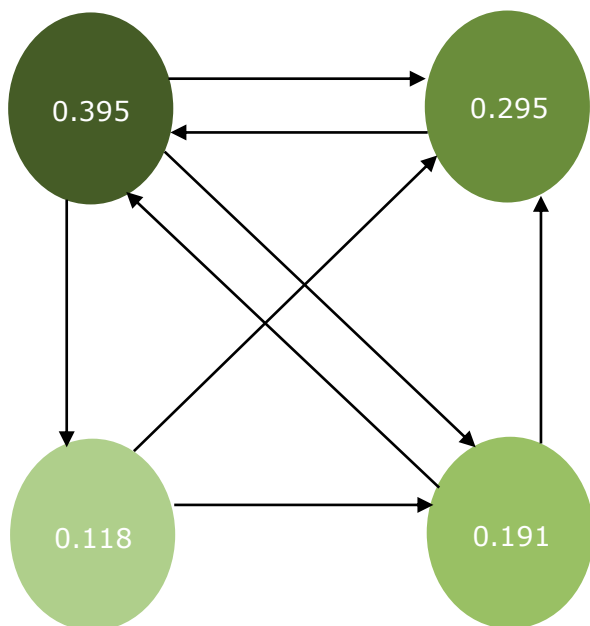
$k=0$



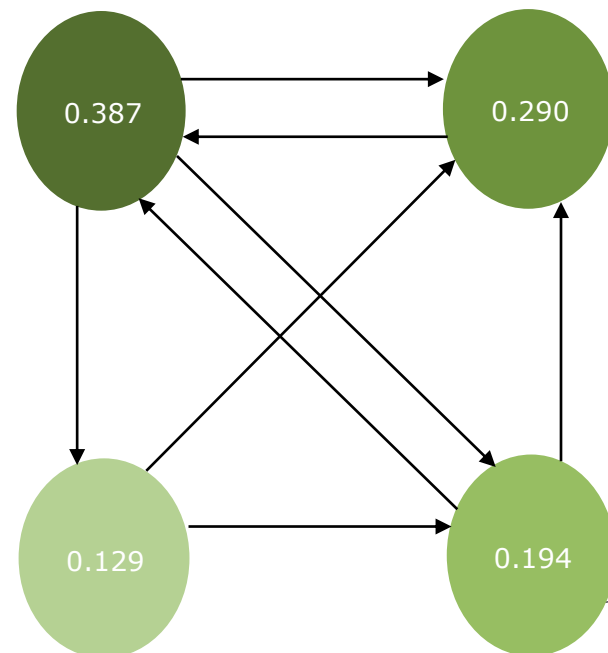
$k=1$



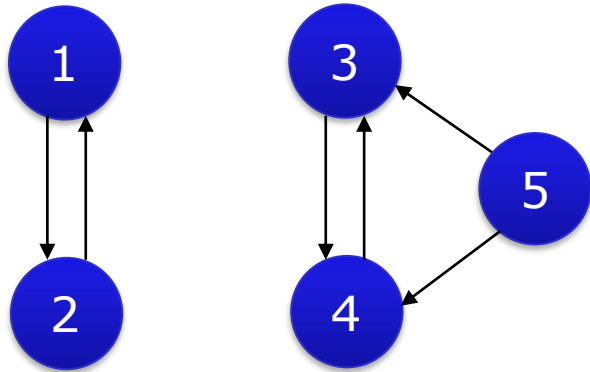
$k=4$



$k \rightarrow \infty$



Possible Problems



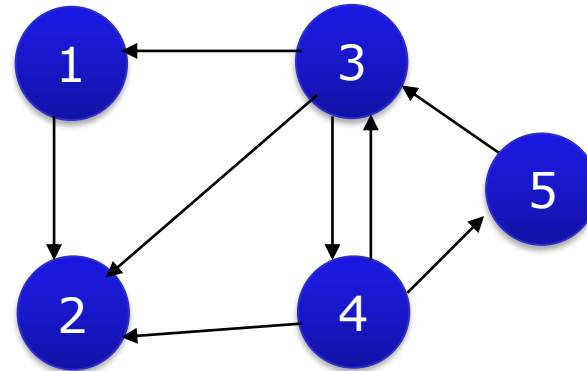
Disconnected graphs

$$M = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/2 \\ 0 & 0 & 1 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{x}^{(1)} = [1/2 \quad 1/2 \quad 0 \quad 0 \quad 0]^T$$

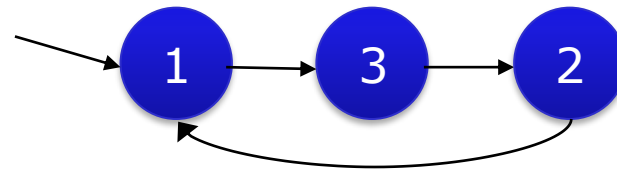
$$\mathbf{x}^{(2)} = [0 \quad 0 \quad 1/2 \quad 1/2 \quad 0]^T$$

and by any linear combination of $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$



Dangling nodes

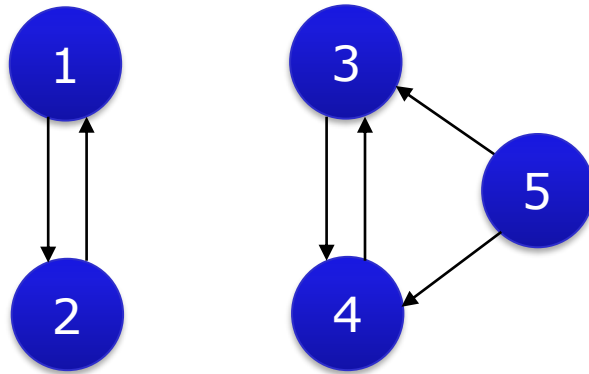
Node 2 has no outgoing links
- e.g. pdf page or a music file



Rank sinks

Some nodes accumulate inflated PageRank scores

Solution: Teleportation (1)



$$M = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/2 \\ 0 & 0 & 1 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G = (1 - \lambda) \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/2 \\ 0 & 0 & 1 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} + \lambda \begin{bmatrix} 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \end{bmatrix}$$

$$G = (1 - \lambda) M + \lambda S$$

G is the google matrix

where $S \in \mathbb{R}^{n \times n}$ $S_{ij} = 1/n$ $0 < \lambda < 1$ **λ is a teleportation parameter** 40

Solution: Teleportation (2)

- For example, setting $\lambda = 0.15$

$$M = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/2 \\ 0 & 0 & 1 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{blue arrow}} G = \begin{bmatrix} 0.03 & 0.88 & 0.03 & 0.03 & 0.03 \\ 0.88 & 0.03 & 0.03 & 0.03 & 0.03 \\ 0.03 & 0.03 & 0.03 & 0.88 & 0.455 \\ 0.03 & 0.03 & 0.88 & 0.03 & 0.455 \\ 0.03 & 0.03 & 0.03 & 0.03 & 0.03 \end{bmatrix}$$

G is the google matrix after teleportation

The unique PageRank score is given by

$$\mathbf{x} = [0.2 \quad 0.2 \quad 0.285 \quad 0.285 \quad 0.03]^T$$

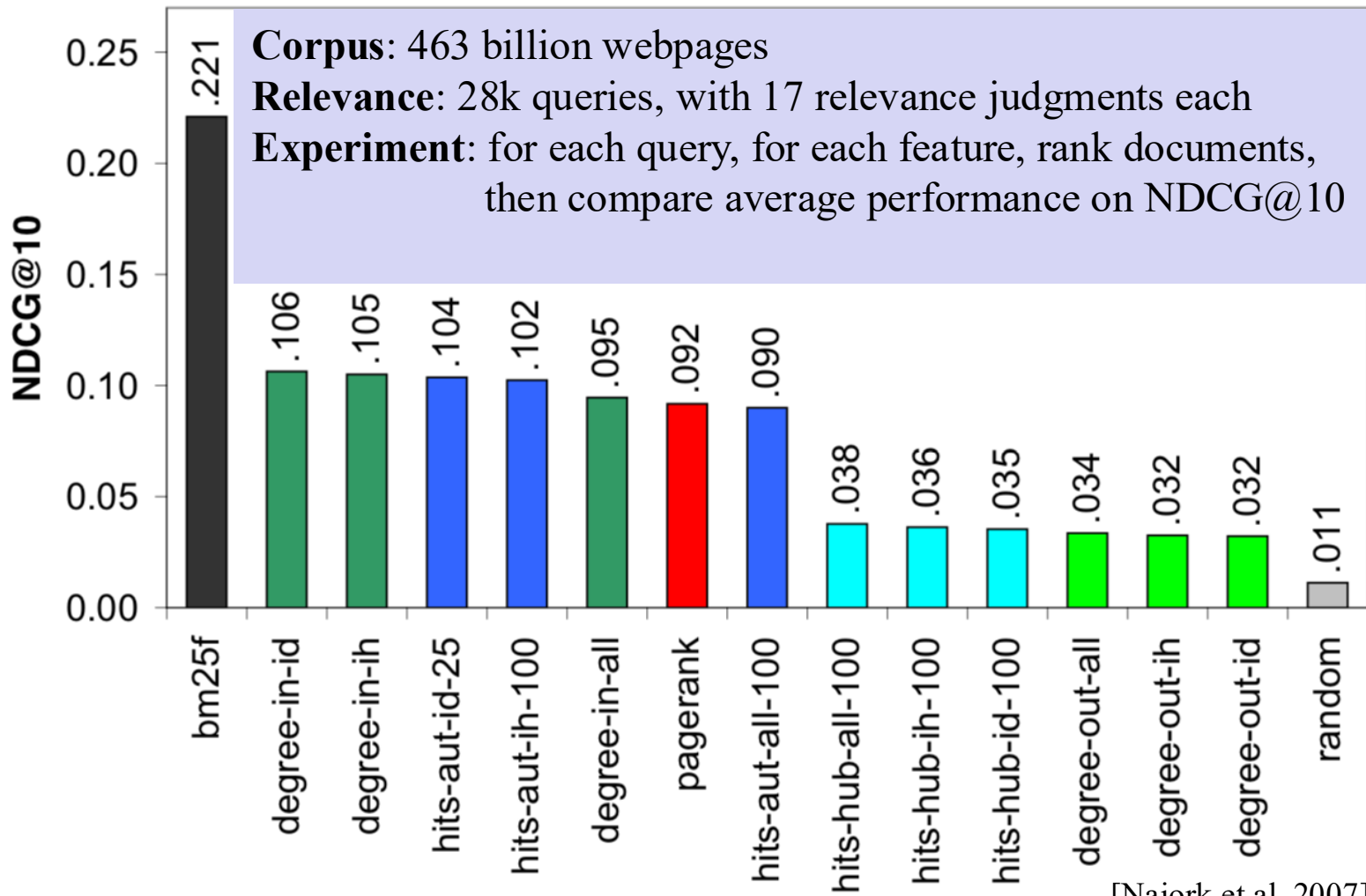
Extensions and Issues

- Extensions
 - Personalized PageRank [Haveliwala 2003]
 - PageRank-directed crawling [Cho and Schonfeld 2007]
 - PageRank without links [Kurland and Lee 2005]
 - Build a non-random surfer [Meiss *et al.* 2010]
- Issues
 - Need a web graph stored in an **efficient** data structure
 - PageRank requires taking powers of a very, **very large matrix**
 - PageRank is an *approximation* of visitation

How Important is PageRank? [Najork et al. 2007]

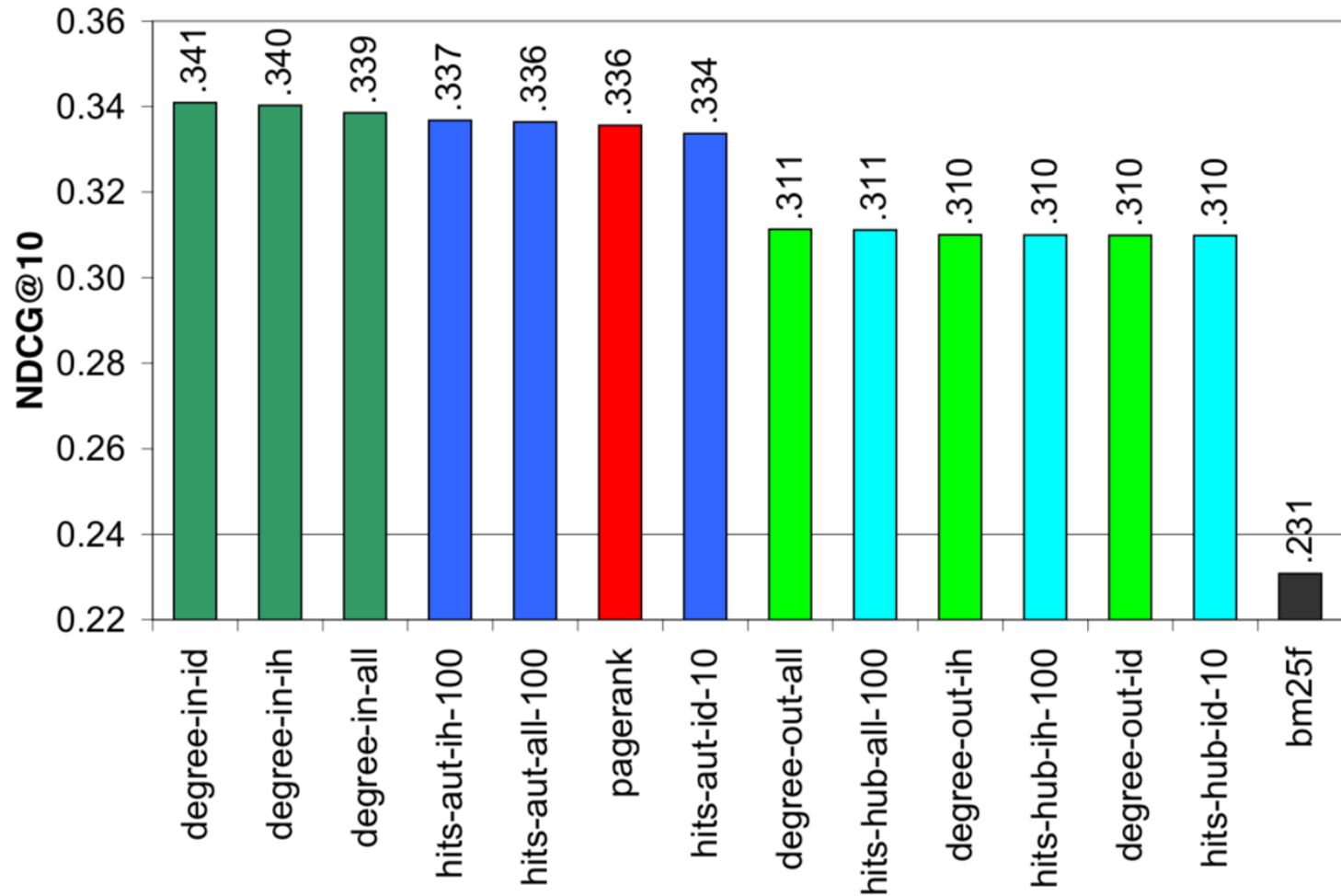
- **Claim:** PageRank is a very important ranking feature reflecting *authority*
- **Test 1:** Compare the empirical performance of PageRank as a ranking feature to:
 - Classic text match features
 - Simpler models (e.g. number of links pointing to a page)
- **Test 2:** Compare the empirical performance of PageRank when *combined* with other ranking signals

Isolated Signals



[Najork et al. 2007]

Signals Combined with BM25F



So, How Important is PageRank?

- Evidence from [Najork et al. 2007] does not support the claim that PageRank is superior to *simpler* models of usage (e.g. in-degree)
- Similar conclusions were reached using TREC web test collections, with no clear benefits in using PageRank
 - In general, link-based ranking features are **weak** features

Other Query Independent Features

- **URL** features: number of slashes in URL, length of URL
- Document **spam** score [Cormack *et al.* 2011]
- Text **quality** score: score combining various text quality features (e.g. readability)
- **Click-through rate**: observed CTR of the page in search results [Dupret and Liao 2010]
- **Dwell** time: average time spent by the users on the page
- Page **load** time: average time it takes to receive the page from its server
- **Social media** links [Dong *et al.* 2010]
- **Neural** Quality Scores [Chang *et al.* 2025]

Query Dependent Features

Information retrieval

From Wikipedia, the free encyclopedia

Information retrieval (IR) is the science of searching for documents, for **information** within documents, and for **metadata** about documents, as well as that of searching relational databases and the World Wide Web. There is overlap in the usage of the terms **data retrieval**, **document retrieval**, **information retrieval**, and **text retrieval**, but each also has its own body of literature, theory, praxis, and technologies. IR is interdisciplinary, based on computer science, mathematics, library science, **information science**, **information architecture**, cognitive psychology, linguistics, and statistics.

Automated **information retrieval** systems are used to reduce what has been called "**information overload**". Many universities and public libraries use IR systems to provide access to books, journals and other documents. Web search engines are the most visible IR applications.

Contents [hide]

- History
 - 1.1 Timeline
- Overview
- Performance measures
 - 3.1 Precision
 - 3.2 Recall
 - 3.3 Fall-Out
 - 3.4 F-measure
 - 3.5 Average precision
 - 3.6 R-Precision
 - 3.7 Mean average precision
 - 3.8 Discounted cumulative gain
 - 3.9 Other Measures
- Model types
 - 4.1 First dimension: mathematical basis
 - 4.2 Second dimension: properties of the model
- Major figures
- Awards in the field
- See also
- References
- External links

History

[edit]

The idea of using computers to search for relevant pieces of **information** was popularized in the article *As We May Think* by Vannevar Bush in 1945.^[1] The first automated **information retrieval** systems were introduced in the 1950s and 1960s. By 1970 several different techniques had been shown to perform well on small text corpora such as the Cranfield collection (several thousand documents).^[2] Large-scale **retrieval** systems, such as the Lockheed Dialog system, came into use early in the 1970s.

In 1992, the US Department of Defense along with the National Institute of Standards and Technology (NIST), cosponsored the Text **Retrieval** Conference (TREC) as part of the TIPSTER text program. The aim of this was to look into the **information retrieval** community by supplying the infrastructure that was needed for evaluation of text **retrieval** methodologies on a very large text collection. This catalyzed research on methods that scale to huge corpora. The introduction of web search engines has boosted the need for very large scale **retrieval** systems even further.

The use of digital methods for storing and retrieving **information** has led to the phenomenon of **digital obsolescence**, where a digital resource ceases to be readable because the physical media, the reader required to read the media, the hardware, or the software that runs on it, is no longer available. The **information** is initially easier to retrieve than if it were on paper, but is then effectively lost.

Timeline

[edit]

- Before the 1900s
 - 1890s: Herman Hollerith invents the recording of data on a machine readable medium.
 - 1890 Hollerith cards, key punches and tabulators used to process the 1890 US Census data.
- 1940s–1950s
 - late 1940s: The US military confronted problems of indexing and **retrieval** of wartime scientific research documents captured from Germans.
 - 1945: Vannevar Bush's *As We May Think* appeared in *Atlantic Monthly*.
 - 1947: Hans Peter Luhn (research engineer at IBM since 1941) began work on a mechanized punch card-based system for searching chemical compounds.

- Vector space model scores
- BM25, PL2, LM scores
- Proximity scores

Using document structure

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- BM25F, PL2F

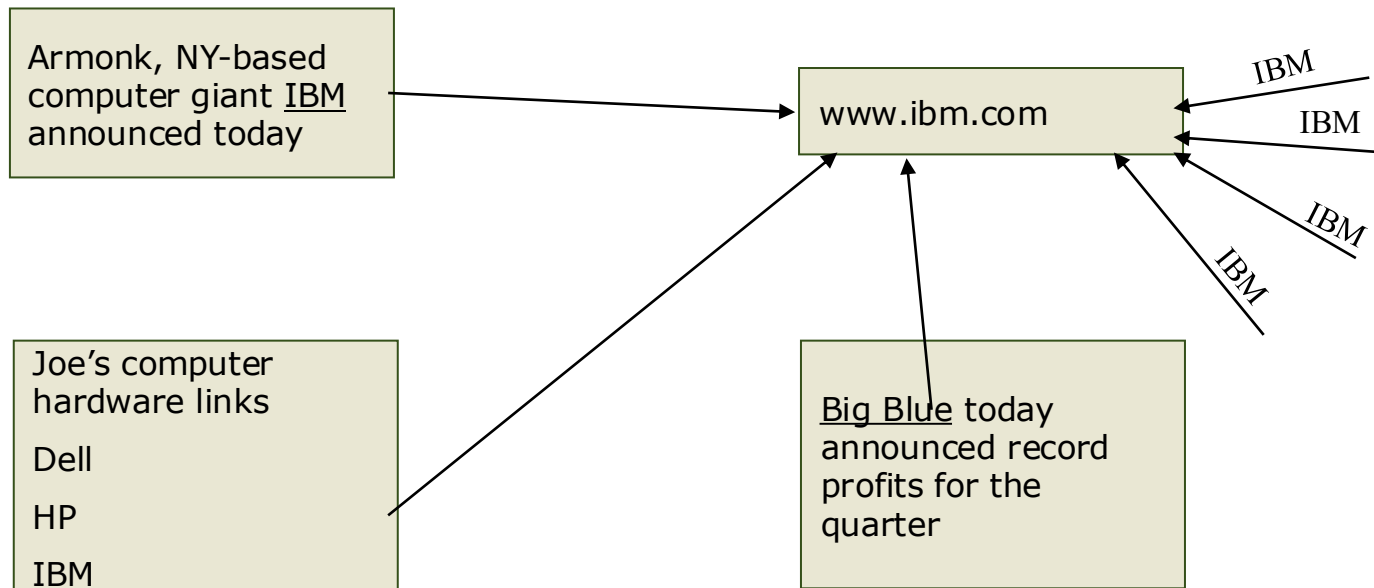
Query Dependent Features



- Include anchor text as a source of text information [Robertson et al. 2004]
- Similar to manual document keyword assignment

Anchor Text Importance

- **Anchor text** tends to be short, descriptive, and similar to query text
- Helps when descriptive text in destination page is embedded in image logos rather than in accessible text
- **Increases content** for popular pages with many in-coming links, increasing recall of these pages



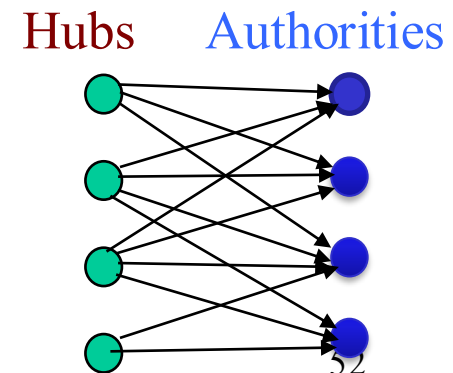
Indexing Anchor Text

- Retrieval experiments have shown that anchor text has **significant impact** on effectiveness for some types of queries
 - i.e., more than PageRank
- However, it can have unexpected **side effects**:
 - Sometimes anchor text is not useful: “click here”
 - Orchestrated campaigns: *e.g.*, “*evil empire*” as in: `<a href=“http://www.google.com”>Evil Empire</a>`
 - IBM’s copyright page (high term freq. for ‘ibm’)
 - Spam links pointing to itself
- **Solution:** Anchor text field can have less weight than the body field

What are the issues in incorporating anchor text as an index feature for web retrieval?

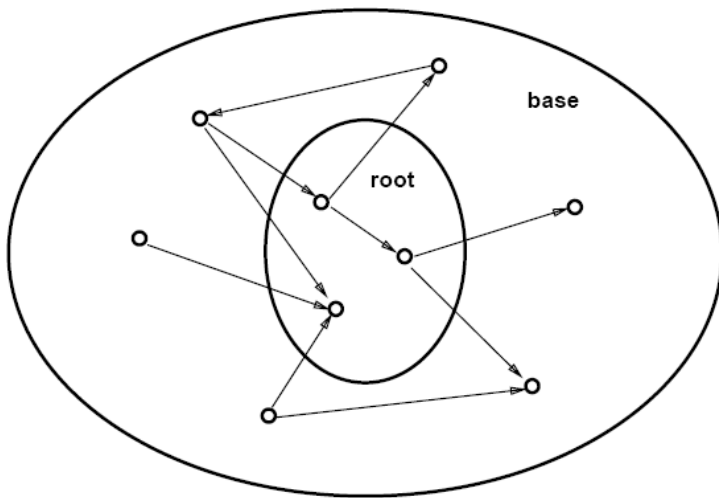
HITS: Hyperlink-Induced Topic Search [Kleinberg 1998]

- A **Query dependent** feature. Link analysis is carried out over a query-induced graph
- In response to a query, instead of a single ranked list of webpages, compute two inter-related scores:
 - **Hub** scores are high for pages that provide lots of useful links to content pages (topic authorities)
 - **Authority** scores are high for pages that have many incoming links from good hubs for the subject
- **Mutually recursive process:** Good **hubs** point to good **authorities**. Good **authorities** are pointed by good **hubs**



HITS Process

1. Send query q to an IR system to obtain the **root set** R of retrieved top- k pages
2. Expand the root set by radius one to obtain an expanded graph S (**base set**)
 - Add pages with **outgoing** links to pages in the root set
 - Add pages with **incoming** links from pages in the root set



3. Maintain for each page $p \in S$

Authority score: a_p (vector $\mathbf{a}$)

Hub score: h_p (vector $\mathbf{h}$)

$$\sum_{p \in S} (a_p)^2 = 1 \quad \sum_{p \in S} (h_p)^2 = 1$$

HITS Iterative Algorithm

Initialize for all $p \in S$: $a_p = h_p = 1$

For $i = 1$ to k :

For all $p \in S$: $a_p = \sum_{q:q \rightarrow p} h_q$ Update authority scores

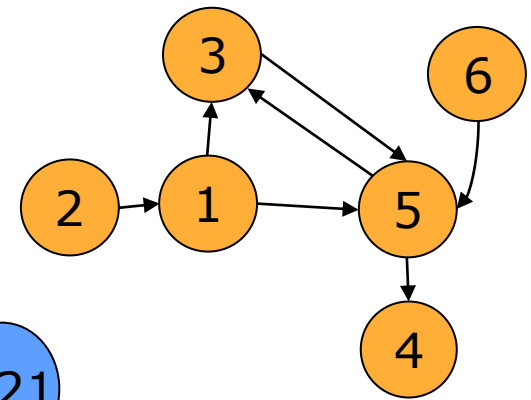
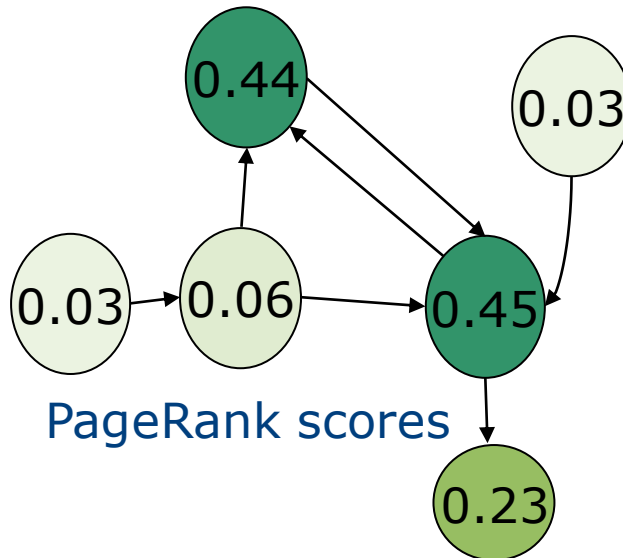
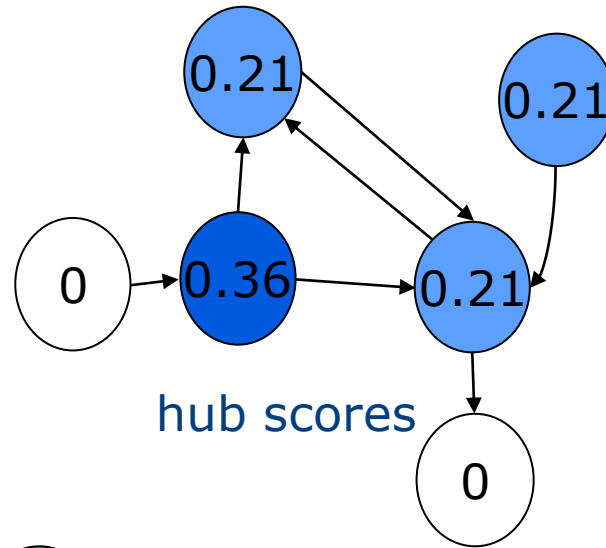
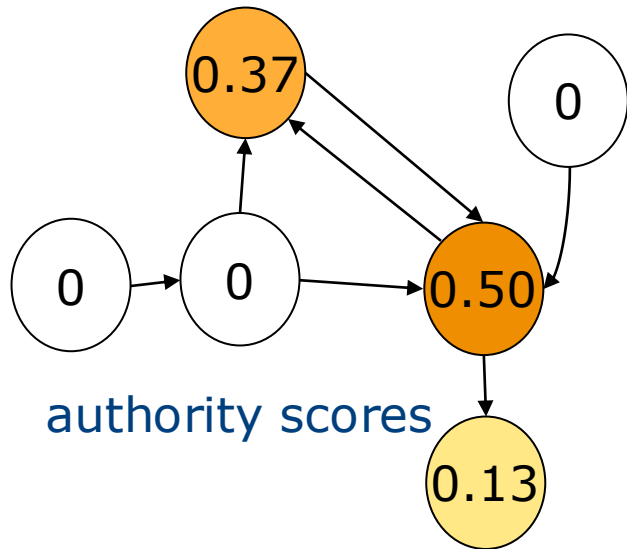
For all $p \in S$: $h_p = \sum_{q:p \rightarrow q} a_q$ Update hub scores

For all $p \in S$: $a_p = a_p / c$ $c: \sum_{p \in S} (a_p / c)^2 = 1$ (normalize $\mathbf{a}$)

For all $p \in S$: $h_p = h_p / c$ $c: \sum_{p \in S} (h_p / c)^2 = 1$ (normalize $\mathbf{h}$)

$\mathbf{a}$ converges to the principal eigenvector of $A^T A$ and $\mathbf{h}$ converges to that of $A A^T$ - A Adjacency Matrix for S : $A_{ij} = 1$ for $i \in S, j \in S$ iff $i \rightarrow j$

Comparative Example



Other Query Dependent Features

- Matches with a user's browsing history [Shen 2005]
- Matches with a user's reading level [Kim *et al.* 2012]
- Number of previous **clicks** on a document for a given query

Query Features

- Number of query words: **query length** often suggests different retrieval strategies
 - **short queries**: navigational (e.g. facebook)
 - **long queries**: topic-based (e.g. ‘tutorial on expectation-maximization algorithms’)
- Trigger words: **certain** words in the query suggest different retrieval strategies
 - e.g. ‘**time** in mumbai’, ‘brooklyn **weather**’

Modern Rankers

- Modern rankers take 1000s of **features** (inc **Neural Net approaches like BERT**)
- PageRank and/or other link analysis-based features are just a few of them
- **Text matching** is not everything
- **Anchor text** can be regarded as reference
- **Clicks** can be considered as votes
- **Context of user** (location, time, etc.) is very important.

Hand-tuning this many features is infeasible but machine learning algorithms can be used to tune a ranking function

Training Data in Learning to Rank

- **Training query sets** (ideally representative of real search logs)
 - Need to sample both prevalent queries in the query logs and less frequent query types
 - The machine learning algorithm will learn only for those types of queries it observes
- Several documents collected for each query via “**pooling**”
 - e.g. take the top m documents from k different ranking features (or systems).
- Query-document pairs are “judged” by **professional/paid editors**
 - Recruit editors who will be able to understand the task and relevance
 - Inappropriate editors = bad data = wasted time/effort
- *(Absolute) relevance judgments* are often multi-graded (e.g. Bad, Good, Perfect)
 - Could be a *preference judgment* (doc_i is more relevant than doc_j for the query)
 - Bad guidelines = bad data = wasted time/effort

Relevance Judgments in TREC

1. **Nav:** This page represents a home page of an entity directly named by the query; the user may be searching for this specific page or site. (*relevance grade 4*)
2. **Key:** This page or site is dedicated to the topic; authoritative and comprehensive, it is worthy of being a top result in a web search engine. (*grade 3*)
3. **HRel:** The content of this page provides substantial information on the topic. (*grade 2*)
4. **Rel:** The content of this page provides some information on the topic, which may be minimal; the relevant information must be on that page, not just promising-looking anchor text pointing to a possibly useful page. (*grade 1*)
5. **Non:** The content of this page does not provide useful information on the topic, but it may provide useful information on other topics, including other interpretations of the same query. (*grade 0*)
6. **Junk:** This page does not appear to be useful for any reasonable purpose; it may be spam or junk. (*grade 0*)

Retrieval Strategy (s)

- There are *many* ways to structure a relationship between input features and relevance scores
 - linear models
 - decision trees
 - support vector machines
 - ...
- Each learning to rank strategy carries a unique set of parameters and methods to learn them
- C.f. Lecture 8 (**Learning to Rank**) for more details on learning to rank methods

Validating the System

- How well the system is doing and/or how well a newly introduced feature is doing?

Baseline Ranking Algorithm

[Personalized Search](#)

Personalized Search ▶ Personalized Web Search Personalized Web ▶ Data Integration in Web Data Extraction System Personalized Web Search J I - R O N G ...
research.microsoft.com/pubs/79334/publishedversion.pdf · PDF file

[A personalized search research based on vocabulary semantic net](#)

Along with the fast developing of network technology, the number of Web page and user of network search become very enormous. In order to solve the problem of ...
portal.acm.org/citation.cfm?id=1794768

[Zakta – Personalized Social Search Engine - edit search, fire](#)

Zakta, unlike other social search engines, can be considered as a "Personalized Search Engine" due to its ability to dig deeper to get the required information and ...
techpp.com/2009/10/15/zakta-personalized-social-search-engine

Related Searches for **personalized search research**

[Ontology-based Personalized Search](#)
[Bing Personalized Search](#)
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[Disable Personalized Search](#)
[Personalized Search Results](#)
[Personalization Business](#)

[Personalized search - Wikipedia, the free encyclopedia](#)

Personalized search refers to search experiences ... specific groups of people, personalized search depends on a user profile that is unique to the individual. Research ...
en.wikipedia.org/wiki/Personalized_Search

[Research from Microsoft: Personalized Search, Determining a Query ...](#)

The other day I posted about a paper presented at the SIGIR conference a few week's ago. Apparently, that got Findory CEO, Greg Linden, looking for other...
blog.searchenginewatch.com/blog/050826-121640

[Adapting SEO for Personalized Search](#)

Ok, but seriously, the last round of personalized search research we did here on it seems to suggest that a lot of the personalization, in relatively new query ...
www.searchenginejournal.com/adapting-seo-for-personalized-search/22207

Proposed Ranking Algorithm

[ACM SIGIR Special Interest Group on Information Retrieval Home Page](#)

Welcome to the ACM SIGIR Web site. ACM SIGIR addresses issues ranging from theory to user demands in the application of computers to the acquisition, organization ...
www.sigir.org

[Personalized Search via Automated Analysis of Interests and ...](#)

Personalized Search via Automated Analysis of Interests and Activities Jaime Teevan MIT, CSAIL 32 ...
Cambridge, MA 02138 USA tee van@csail. mit. edu Susan T ...
research.microsoft.com/en-us/um/people/sdumais/SIGIR2005-PersonalizedSearch.pdf · PDF file

[Personalized search](#)

Personalized search refers to search experiences ... Research systems that personalize search results model their users in ... to personalize global Web search". SIGIR '08 Proceedings of the ...
research.microsoft.com/en-us/um/people/sdumais

[Research page](#)

Research page I am interested ... issues, including: personal information management, web search ... and ... perspective. SIGIR 2010 Desktop Search Workshop ...
research.microsoft.com/en-us/um/people/sdumais

[Personalized search - Wikipedia, the free encyclopedia](#)

Personalized search refers to search experiences ... Research systems that personalize search results model their users in ... to personalize global Web search". SIGIR: 287 ...
en.wikipedia.org/wiki/Personalized_Search

[Xuehua's Publications](#)

Proceedings of 2003 ACM Conference on Research and Development on Information Retrieval (SIGIR2003), pages 377-378. pdf ppt; Demos. UCAIR Toolbar: A Personalized Search ...
sifaka.cs.uiuc.edu/xshen/publication.html

[Event: IR](#)

SIGIR is the major international forum for the presentation of new research results and for the demonstration of ... summarization, task models, personalized search ...
portal.acm.org/browse_dl.cfm?linked=1&part=series&idx=SERIES278&coll=ACM&dl=ACM

Which is better?

See Evaluation (Lecture 5, Lecture 11)